3 **(a)** Define *gravitational potential* at a point.

……

…… [1]

(b) Fig. 3.1 shows three equipotential surface centred about the Earth, each with its value indicated.

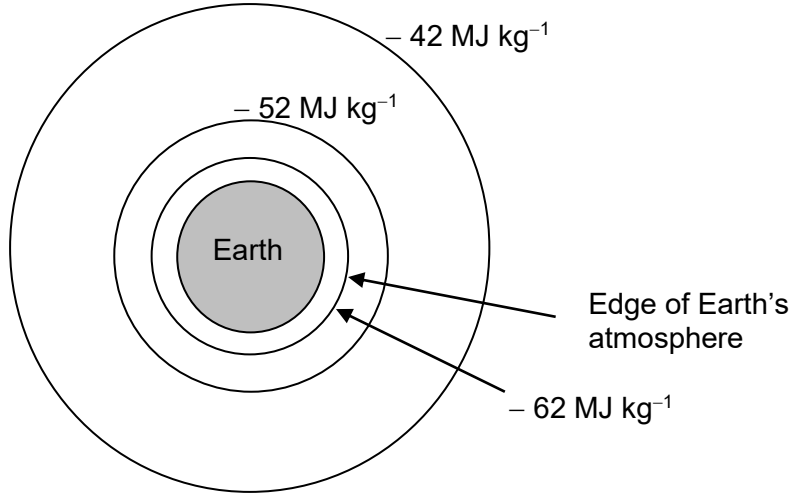

Fig. 3.1

(i) Explain why

1. the spacing between equipotential surfaces are not the same,

……

…… [1]

2. the values of equipotential is negative.

……

……

……

…… [2]

(ii) The potential at the Moon's orbit around the Earth is -1.0 MJ kg^{-1}.
If a spacecraft uses only the gravitational attraction of the Earth to travel from the Moon to the top of the Earth's atmosphere, determine its arrival speed.

arrival speed = …….. m s^{-1} [3]

(iii) The gravitational potential at the edge of the Earth's atmosphere is $-$ 61.6 MJ kg^{-1}.
If the radius of the Earth is 6371 km and its mass is 5.97×10^{24} kg , determine the height of this edge of the atmosphere from the Earth's surface.

height = ……. m [3]